

Predicting Psychiatric Illness Risk from Neuroimaging–Genetics Associations

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Using SNP associations to brain structure as an intermediate to predict psychiatric disease risk

The problem

Psychiatric disorders are highly polygenic. Hundreds to thousands of common variants (SNPs) each contribute a tiny effect — there is no single “disease gene”. This makes risk hard to predict directly from genetics.

Our idea. A SNP may act on illness *through brain structure*. We therefore use a SNP’s measured associations to $\sim 1,000$ MRI-derived brain phenotypes as features, and ask whether they predict that SNP’s association to the illness.

Central question:

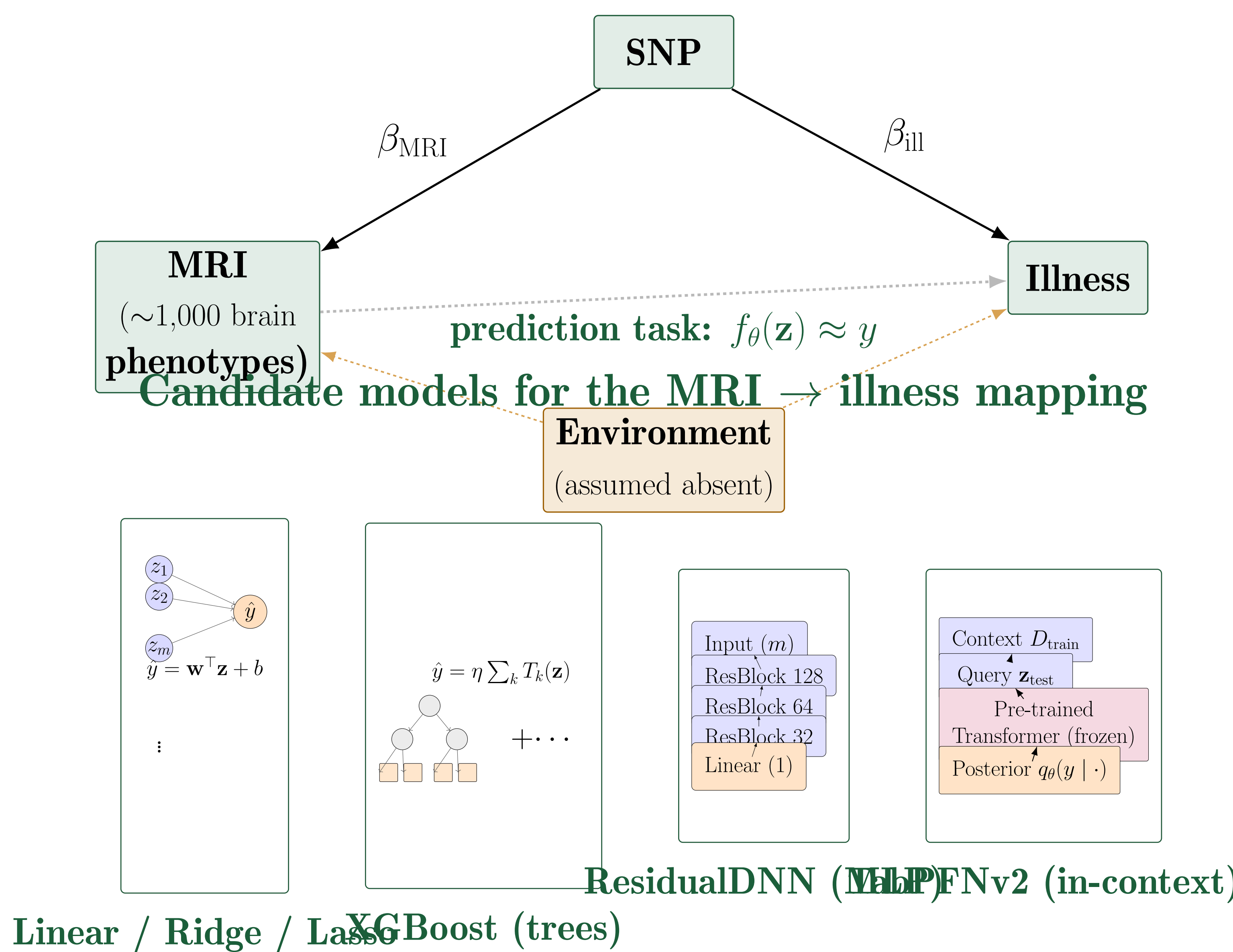
Can MRI-derived genetic associations predict the SNP–illness association?

What we hope to gain

- Identify **neurobiological biomarkers** — the specific brain regions / phenotypes that carry illness signal.
- Test whether the MRI $\rightarrow$ illness link is **non-linear** rather than purely linear.
- Build a **structured pathway** from genetics to disease, rather than a black-box genotype $\rightarrow$ phenotype predictor.

Key caveat — environment. We currently assume no confounding by environmental factors. A SNP could influence illness through a pathway independent of brain structure; this is important when interpreting any result.

From genetics to illness through the brain — and the models that learn the link



Each model learns $f_{\theta} : \mathbb{R}^m \rightarrow \mathbb{R}$, mapping a SNP’s MRI z-score profile $\mathbf{z}$ to its illness z-score y , fit by mean-squared-error empirical risk minimisation.

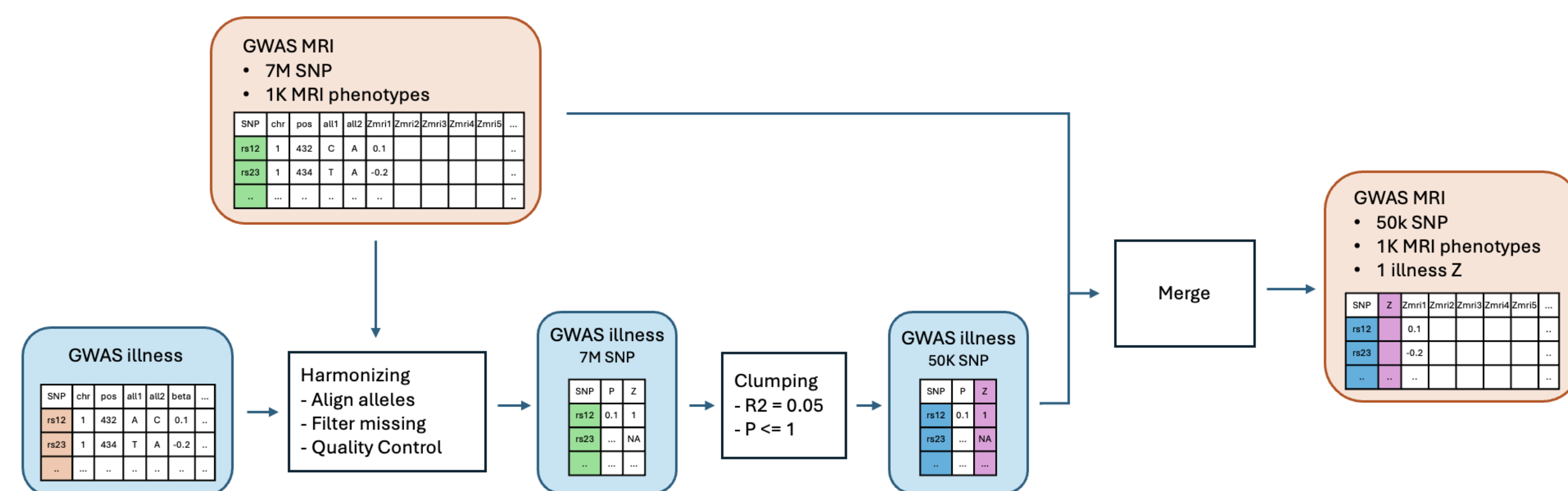
The dataset — source GWAS

Inputs are publicly available **GWAS summary statistics**. For SNP i and phenotype j we use the z-score $z_{i,j} = \beta_{i,j} / \text{s.e.}(\beta_{i,j})$. All cohorts are of European ancestry.

Phenotype Consortium		N_{case}	N_{control}
<i>MRI: $\approx 1,000$ brain features (UK Biobank)</i>			
SCZ	PGC-SWG	53,386	77,258
BIP	PGC	59,287	781,022
MDD	PGC-MDD	59,851	113,154
ADHD	PGC-iPSYCH	38,691	186,843
AZ	IGAP-UKB	71,880	383,378
OCD	PGC-OCD	23,493	1,114,613

SCZ has the largest GWAS $\Rightarrow$ strongest signal, and is our primary testbed.

Construction pipeline (per illness)



Steps: harmonise alleles across cohorts $\rightarrow$ **LD-clump** ($R^2 < 0.05$) to keep independent SNPs $\rightarrow$ merge MRI and illness GWAS on shared lead SNPs.

Result: one tabular dataset per illness. Rows = SNPs, features = MRI z-scores, target = illness z-score:

$$\begin{bmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,m} & y_1 \\ z_{2,1} & z_{2,2} & \cdots & z_{2,m} & y_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{N,1} & z_{N,2} & \cdots & z_{N,m} & y_N \end{bmatrix} \quad m \approx 1,000$$

Experiment design — a 2×2 grid over the causal graph

The SNP is the common cause; each arrow carries its own GWAS p -value. Two filters — on the **MRI** arrow and on the **illness** arrow — cross into four datasets, each isolating a different signal/noise mix.

No MRI filter

Filter on MRI p -value